

Equipment & Connectivity Management Platform (ECM)

A Socio-Technical Approach to Institutional Resource Allocation

Daniel Martínez Gil · Dylan Silva Orrego · Juan Camilo Triana Paipa

Dept. of Computer Engineering · Universidad Distrital Francisco José de Caldas · Bogotá, Colombia



UNIVERSIDAD DISTRITAL
FRANCISCO JOSÉ DE CALDAS
Acreditación Institucional en Alta Calidad

1. INTRODUCTION

Universities in developing contexts face persistent digital divides due to fragmented, manual management of shared technological assets, producing chronic inefficiencies in resource allocation and device tracking.

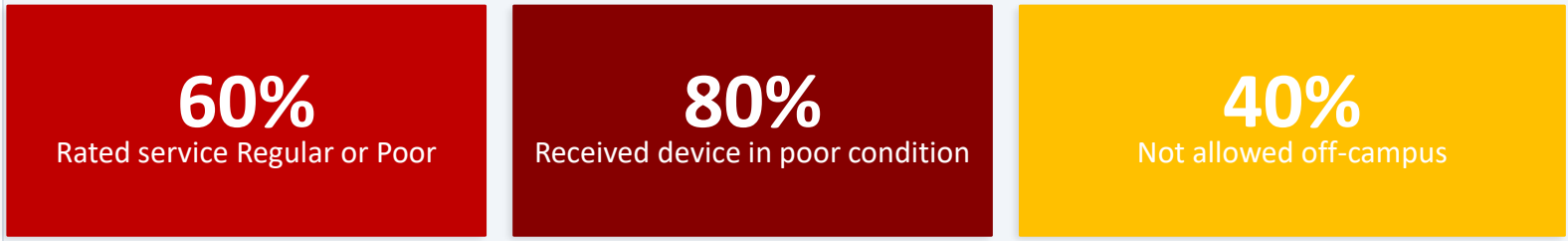


Fig. 1 — Survey results from university engineering students (n = 7,000).

Problem: Inventory inaccuracies · Inefficient allocation · Poor device tracking · Delayed maintenance · Unequal access to technology for ~7,000 users.

Digital divide motivates a centralized automated management system.

2. GOAL

Research Question
Can a socio-technical platform improve allocation, monitoring, and recovery of institutional technological resources while ensuring fairness and scalability?

Expected Outcome: Automated ECM platform supporting 7,000 users via intelligent allocation, risk mitigation, and scalable architecture.

3. PROPOSED SOLUTION

The ECM platform centralizes equipment lending, connectivity management, maintenance tracking, and resource recovery within a Clean Architecture framework.

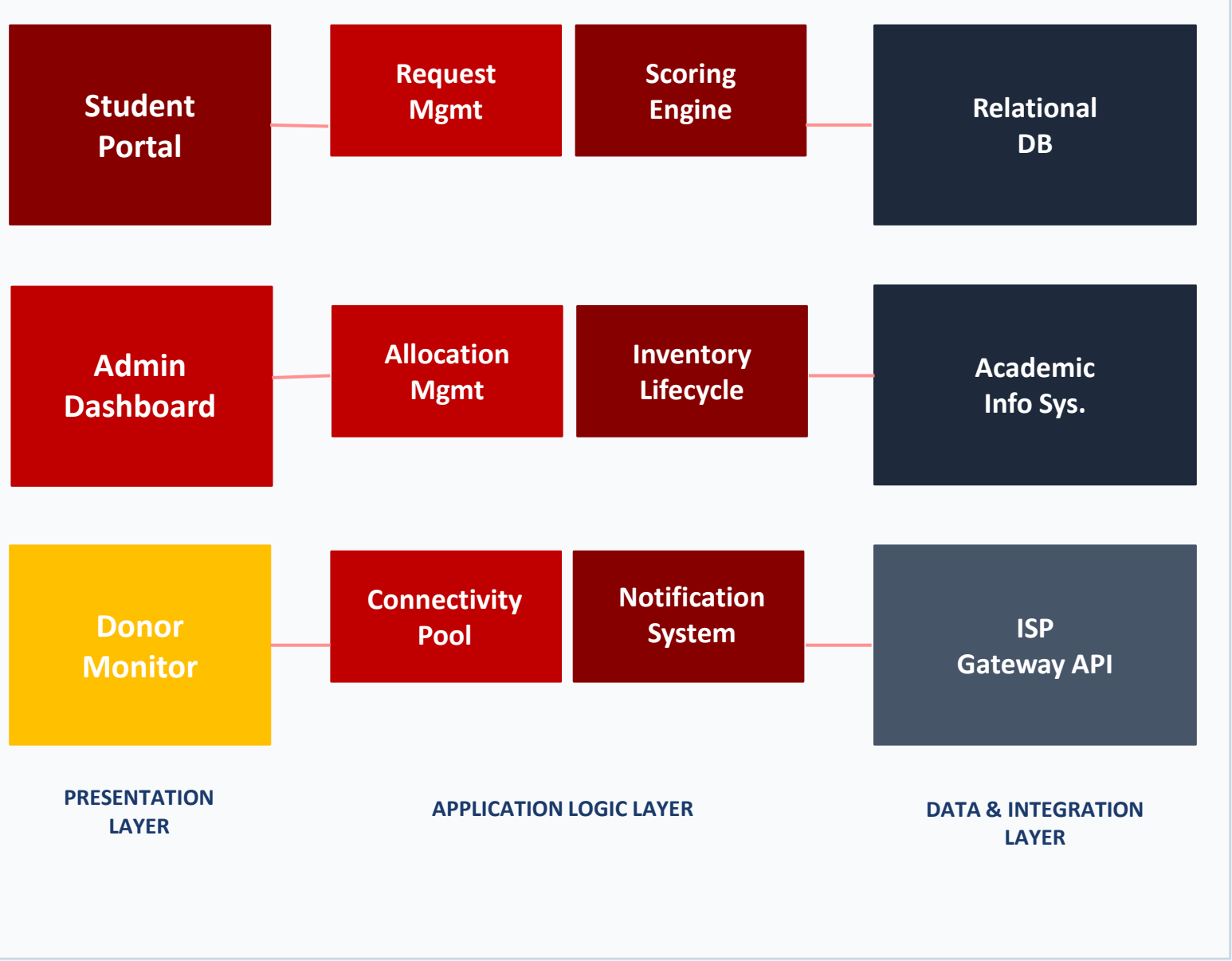


Fig. 2 — ECM Clean Architecture: Three-layer system design.

4. AUTOMATED SCORING ENGINE

$$S = (V \times 0.5) + (A \times 0.3) + (H \times 0.2)$$

V = Vulnerability Index (0.5) A = Academic Load (0.3) H = Historical Compliance (0.2)

Vulnerability (V)

50%

Academic Load (A)

30%

Compliance (H)

20%

Purpose: Prioritize students based on need rather than request order — equity over FIFO.

5. RISK MANAGEMENT & QUALITY ASSURANCE

Risk Mitigation

Academic Hold
Auto-hold on student record for overdue devices

ISP Fail-Safe Cache
Emergency connectivity tokens during outages

Monthly Recalibration
Scoring weights adapt to demand shifts

QA Framework

Tier 1
Unit Tests
> 80% Coverage

Tier 2
Integration Tests
API Contracts

Tier 3
UAT Pilot
< 3 min / request

6. EXPERIMENTS

DES
Discrete Event Simulation

Request → Validation → Scoring → Allocation → Return

ABM
Agent-Based Modeling

7,000 Students · 200 Devices · 1 Admin
180-day simulation horizon

Parameter	Value
Total Users	7,000
Device Fleet	200 units
Demand Rate	12% (~840 req.)
Loan Period	30 days
Simulation Horizon	180 days
Stress Load	1,400 requests

Table 1 — Simulation parameters calibrated from Workshop 1 survey data.

7. RESULTS

Metric	Baseline	Optimized	Stress
Avg. Wait Time (days)	2.1	1.4	8.6
Asset Rotation (days)	9.2	7.8	14.5
Recovery Rate (%)	94.3	98.7	81.2
Scoring Engine Load	840	840	1,400

Table 2 — KPI summary across DES simulation scenarios.

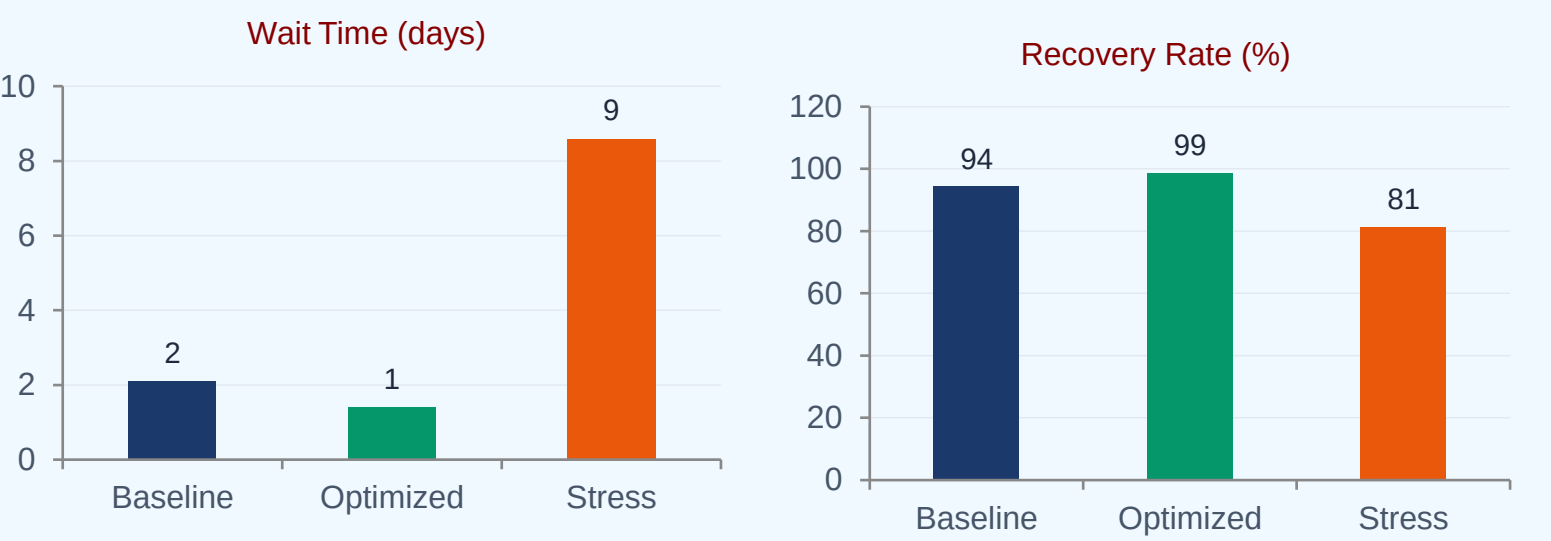


Fig. 3 — DES scenario comparison: wait time (left) and recovery rate (right).

ABM Emergent Behaviors

Demand Cascades	Maintenance Avalanche
67% surge in requests at exam periods	> 25% of fleet in repair simultaneously
Connectivity Bypass	Weight Sensitivity
12% of agents bypass domain filters	V < 0.35 causes equity degradation

8. DESIGN VALIDATION

Scoring Engine Eq.(1)

Academic Hold

ISP Fail-Safe Cache

Weight Recalibration

99.9% Uptime SLA

~34% wait vs. FIFO

Recovery: 85% → 98.7%

0 interruptions in ISP drops

Inequity index ~18%

Degrades > 1,200 concurrent req.

● Confirmed (4/5) ● Partial — scaling required above 1,200 req.

9. CONCLUSIONS

Equitable allocation via Automated Scoring Engine ($S = V \cdot 0.5 + A \cdot 0.3 + H \cdot 0.2$)

Recovery rate above 98.7% with Academic Hold mechanism enabled

34% reduction in average wait time vs. first-come first-served

15% improvement in asset rotation efficiency under optimized scenario

Scaling requirement identified: horizontal auto-scaling above 1,200 concurrent requests

Maintenance avalanche risk: cap concurrent repairs at ≤ 10% of fleet

ECM demonstrates Clean Architecture + Automated Scoring + simulation deliver a scalable, equitable institutional platform ready for pilot deployment.

[1] Evans, Domain-Driven Design. Addison-Wesley, 2003. [2] Martin, Clean Architecture. Prentice Hall, 2017. [3] Banks et al., Discrete-Event System Simulation, 5th ed. Pearson, 2010. [4] Macal & North, Journal of Simulation, vol. 4, 2010. [5] Wilensky & Rand, Agent-Based Modeling. MIT Press, 2015. [6] ISO/IEC 25010:2011. [7] ISO 31000:2018.